

Cleaning a Simulated Global Temperature Record

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I. INTRODUCTION

The supplied teaching dataset contains simulated monthly global temperature anomalies from January 1880 through December 2025. I built a reproducible Python pipeline to repair its deliberately corrupted entries, complete the monthly record, and calculate normalized values. Fig. 1 shows the cleaned series with temperature encoded by both position and color.

II. DATA CLEANING

The pipeline read every field as text, repaired 32 swapped rows, parsed 1,707 data rows, and left no data row unparsed; four trailing non-data lines were discarded. Sorting and retaining the first record per month removed 20 duplicates. The quartiles were $Q_1 = -0.385000$ °C and $Q_3 = 0.490000$ °C, giving an IQR of 0.875000 °C and fences of $[-1.697500, 1.802500]$ °C. The rule removed 63 outliers, all of which were ± 500 or ± 999 sensor codes; it removed no plausible reading. Reindexing produced all 1,752 months, and linear time interpolation filled 190 months: 65 absent months and 125 missing values in existing rows.

III. NORMALIZATION

For each cleaned monthly anomaly x , I calculated

$$d = x - \mu_{20}, \quad (1)$$

and

$$z = \frac{x - \mu}{\sigma}. \quad (2)$$

Here x is the cleaned anomaly in °C, μ_{20} is its mean over 1901–2000, and d is the difference from that baseline used for the color channel. The symbols μ and σ are the mean and population standard deviation over 1880–2025, and z is the standardized anomaly. The calculated values were $\mu_{20} = 0.000159$ °C, $\mu = 0.046434$ °C, and $\sigma = 0.519783$ °C.

IV. RESULTS

Calendar-year means used all 12 cleaned monthly values, including interpolated months. Table I lists the five largest annual means.

TABLE I
FIVE WARMEST YEARS IN THE CLEANED SERIES

Rank	Year	Mean anomaly (°C)	Mean z
1	2024	1.168	2.158
2	2023	1.153	2.130
3	2025	1.013	1.859
4	2016	0.982	1.799
5	2022	0.960	1.759

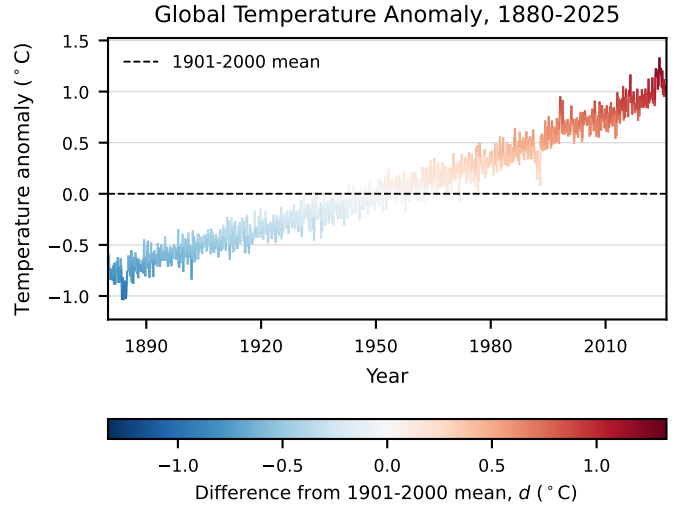


Fig. 1. Cleaned monthly temperature anomaly. Vertical position shows x , while segment color shows d : blue is below and red is above the 1901–2000 mean. The dashed line marks that baseline.